

NAME: _____

CLASS: 11MTX__

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2019

YEAR 11 AP1 EXAMINATION

MATHEMATICS EXTENSION 1

Time allowed – 1 hour plus 5 minutes reading time

DIRECTIONS TO CANDIDATES:

- Write your name on the front page of this question/answer booklet.
- Attempt all questions. Marks are indicated to the right of each question.
- Answer in the space below each question or use the additional writing space provided at the back of this booklet, ensuring all answers are clearly labelled with the question number.
- All necessary working should be shown in every question, other than multiple choice.
- Full marks may not be awarded for careless, badly arranged or badly written work.
- NESA approved calculators may be used. You may bring one handwritten A4 reference sheet into the test.

Topic	Total
1 – Graphical relationships	/ 12
2 – Inequalities	/ 10
3 – Inverse functions	/ 12
4 – Factor and remainder theorems	/ 10
Total	/ 44

Question 1 **F1.1 Graphical relationships**

12 marks

(a) If $f(x) = x^2$ and $g(x) = x^2 + 1$, what is the domain of $f + g$?

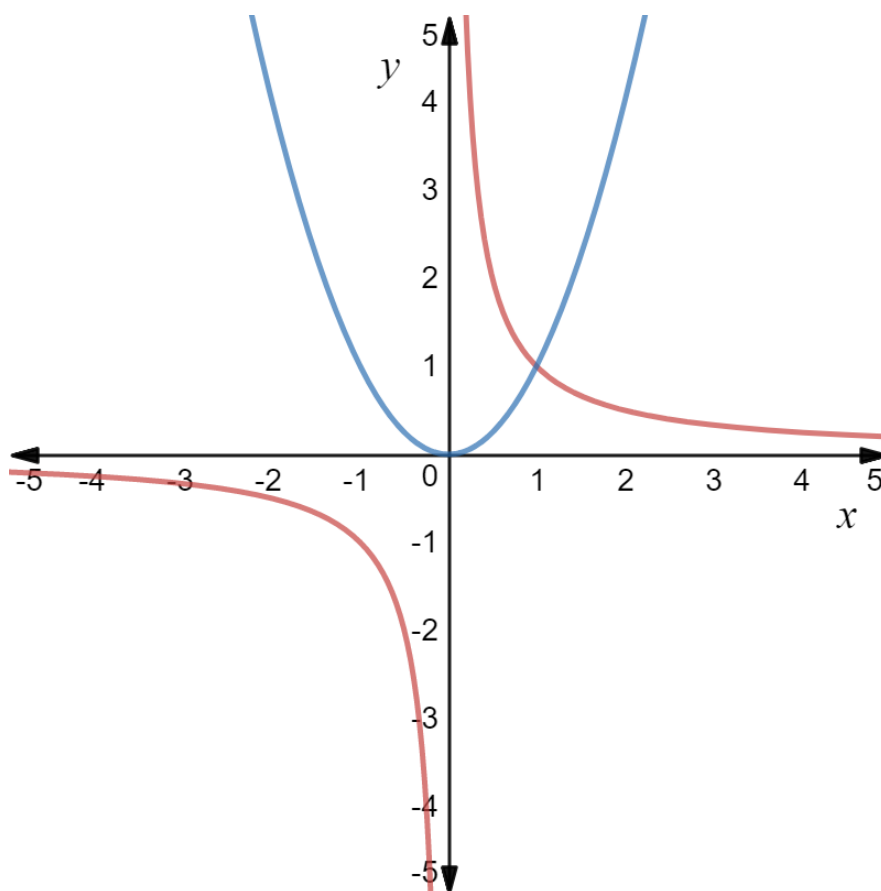
1

Circle the correct answer.

- A. all real x
- B. $x \geq 0$
- C. $x \leq 0$
- D. $x > 0$

(b) Below are the graphs of $f(x) = x^2$ and $g(x) = \frac{1}{x}$

1

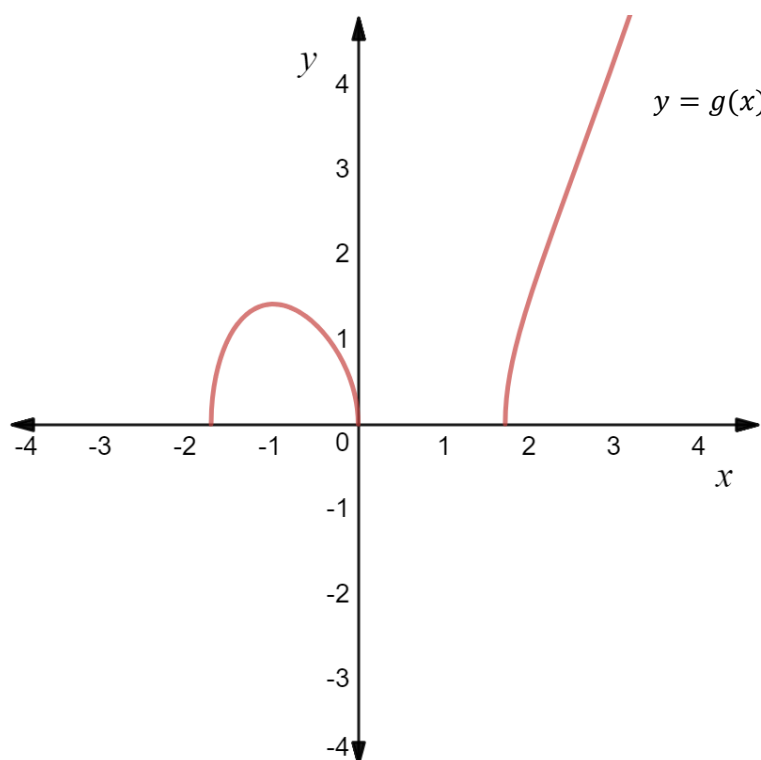
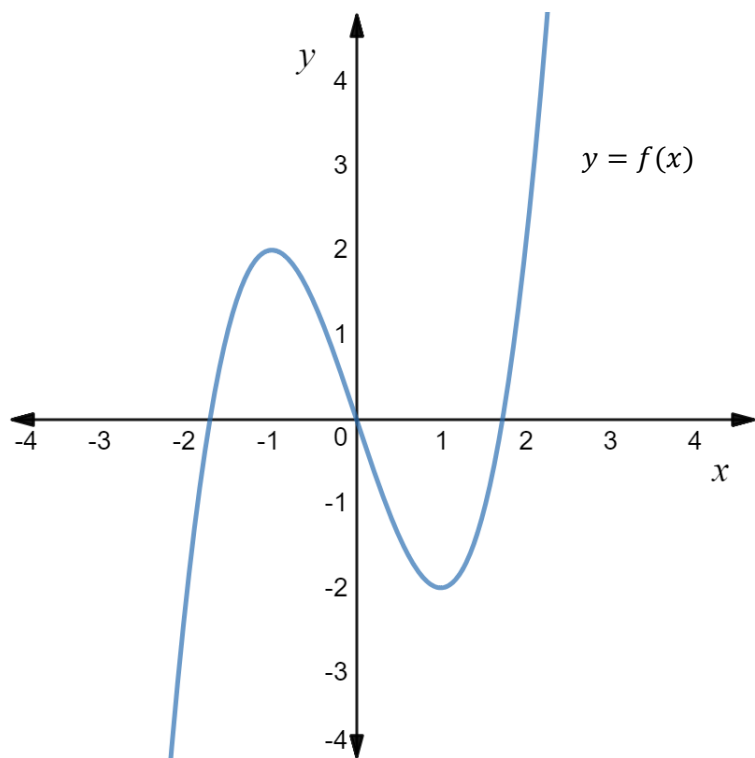


Circle the correct answer which describes the **range** of $f + g$.

- A. the same as the range of f
- B. the same as the range of g
- C. the intersection of the two ranges.
- D. None of the above.

Question 1 continued

(c) Below is a graph and a new function Which equation describes ?



Circle the correct answer:

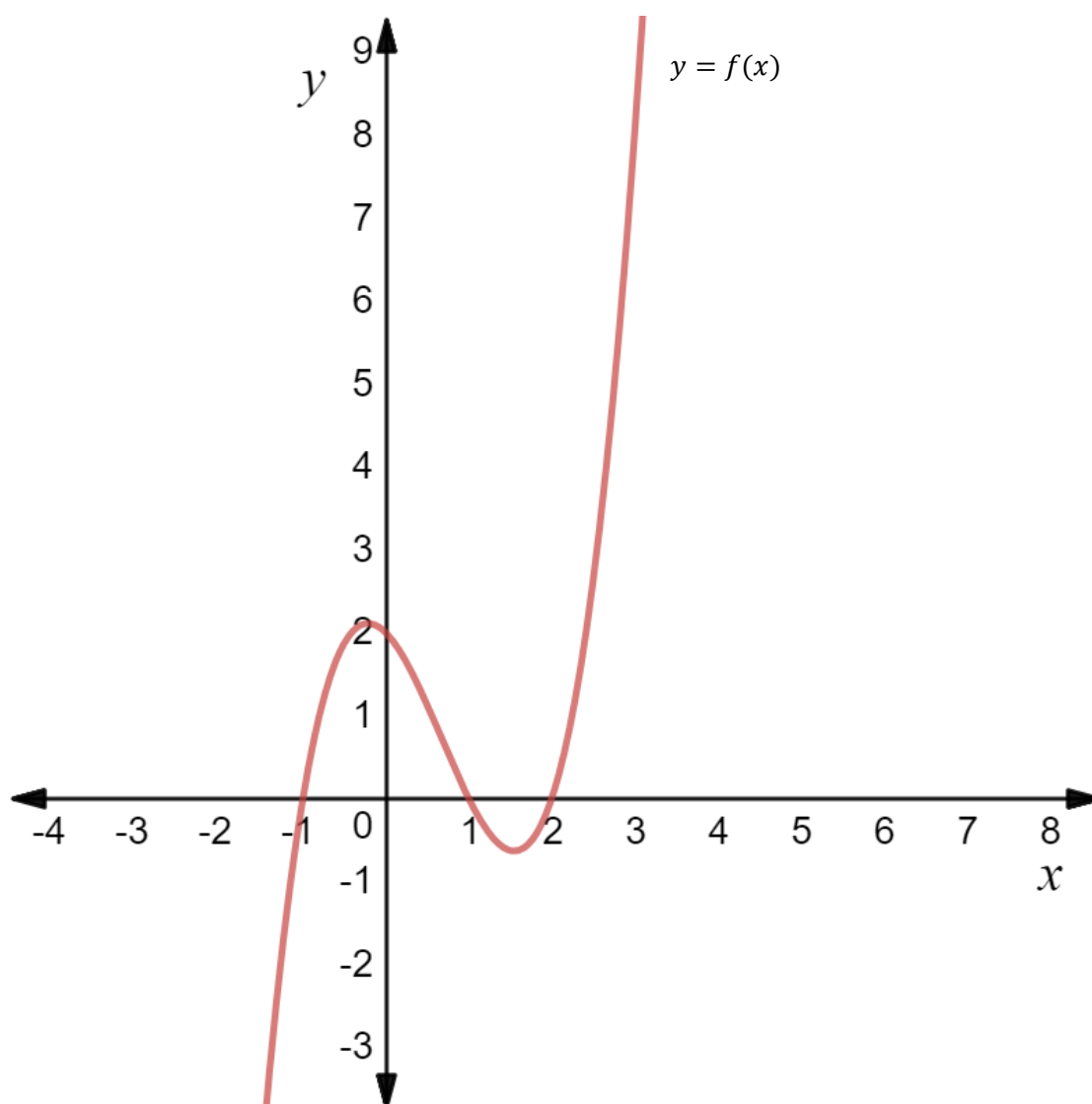
- A.
- B.
- C.
- D.

Question 1 continued

- (d) Describe in words **OR** illustrate with a suitable diagram, how you would draw the graph of $y = f(x)$ if given the graph of $y = g(x)$. 1

- (e) Below is the graph of a function $y = f(x)$. 2

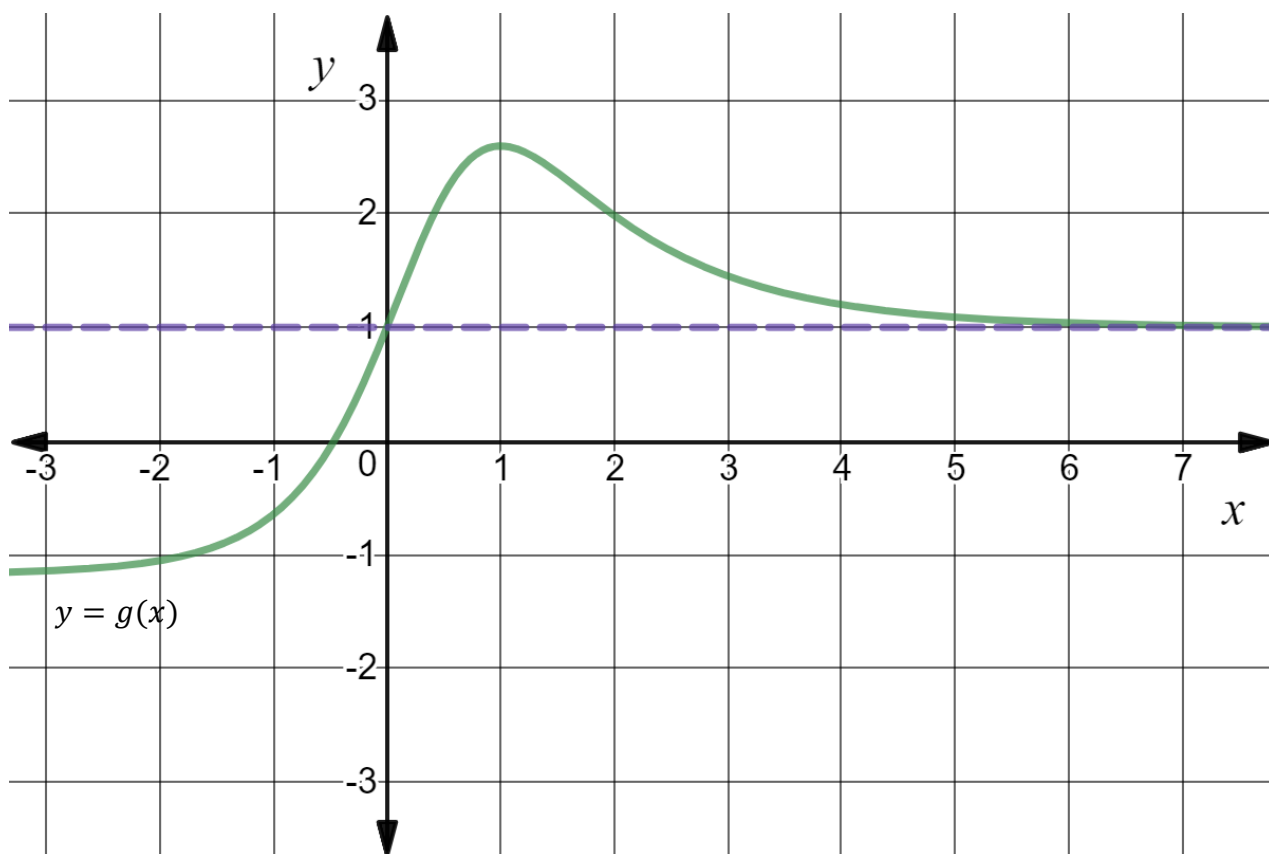
Sketch the graph on the same number plane.



Question 1 continued

- (f) Below is the graph of a function .
Sketch the graph on the same number plane, clearly marking all intercepts and asymptotes.

3

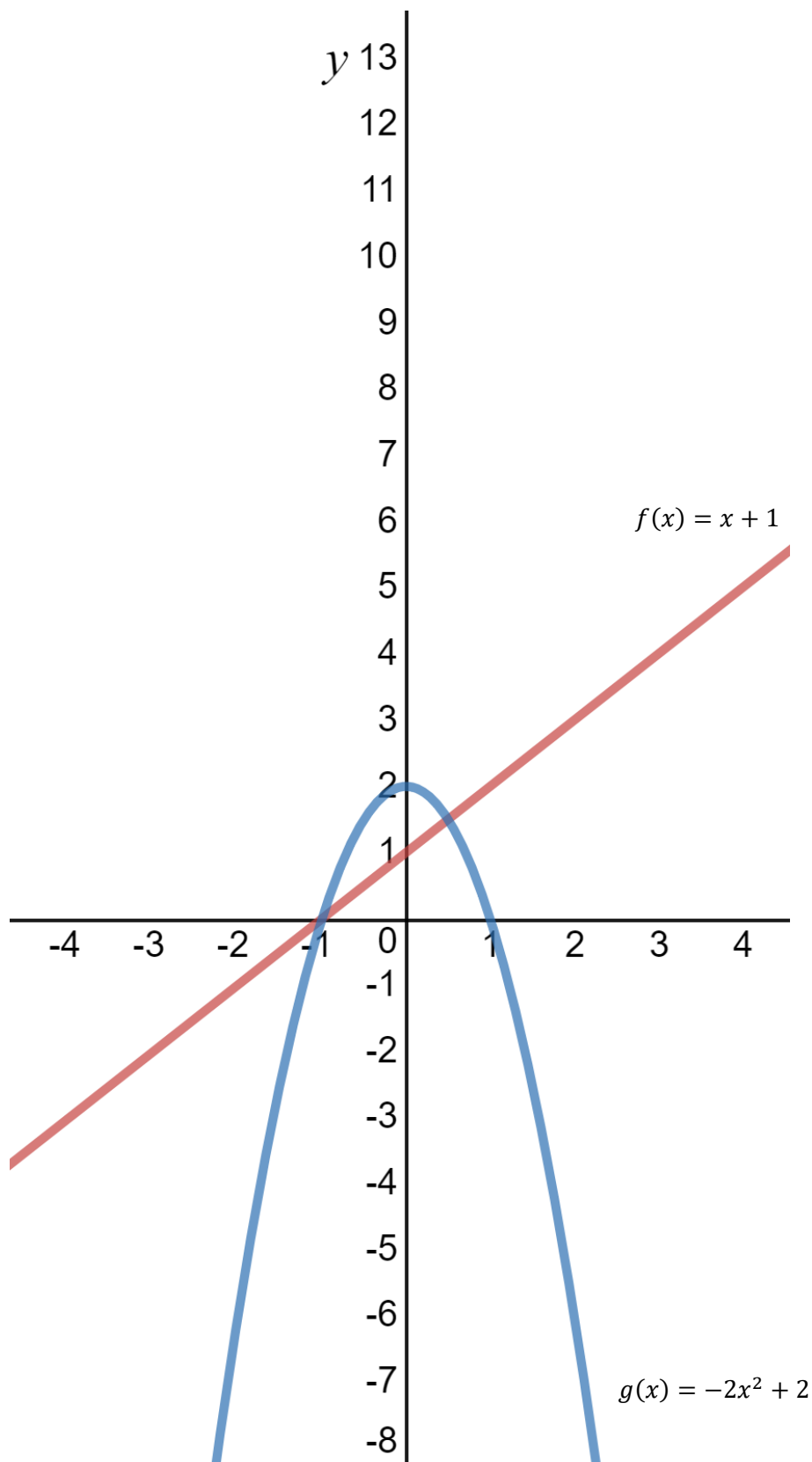


Question 1 continued

(g) On the number plane below are the graphs of two functions and .

3

Sketch the graph of on the same number plane marking all key features.



Question 2**F1.2 - Inequalities****10 marks**

(a) Solve the inequation . Circle the correct answer.

1

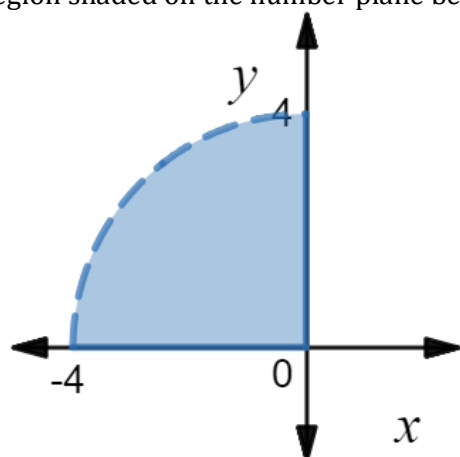
A.

B.

C.

D. No solution

(b) The region shaded on the number plane below is described by which inequalities?

1

Circle the correct answer

A.

B.

C.

D.

(c) Solve .

2

Question 2 continued

(d) Solve and sketch your solution on the number line below.

3

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. At the bottom center, there is a large, black, double-headed arrow pointing left and right, spanning most of the page's width. This graphic is commonly used to represent a number line or a range of values.

(e) Solve

3

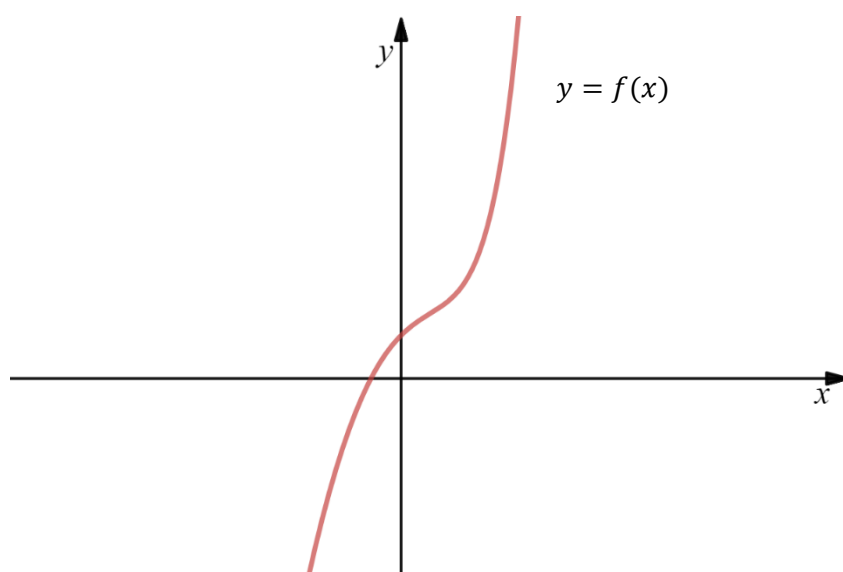
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 3**F1.3 - Inverse Functions****12 marks**

- (a)** If find the equation of the inverse function .

1

- (b)** Below is the graph of a function

1

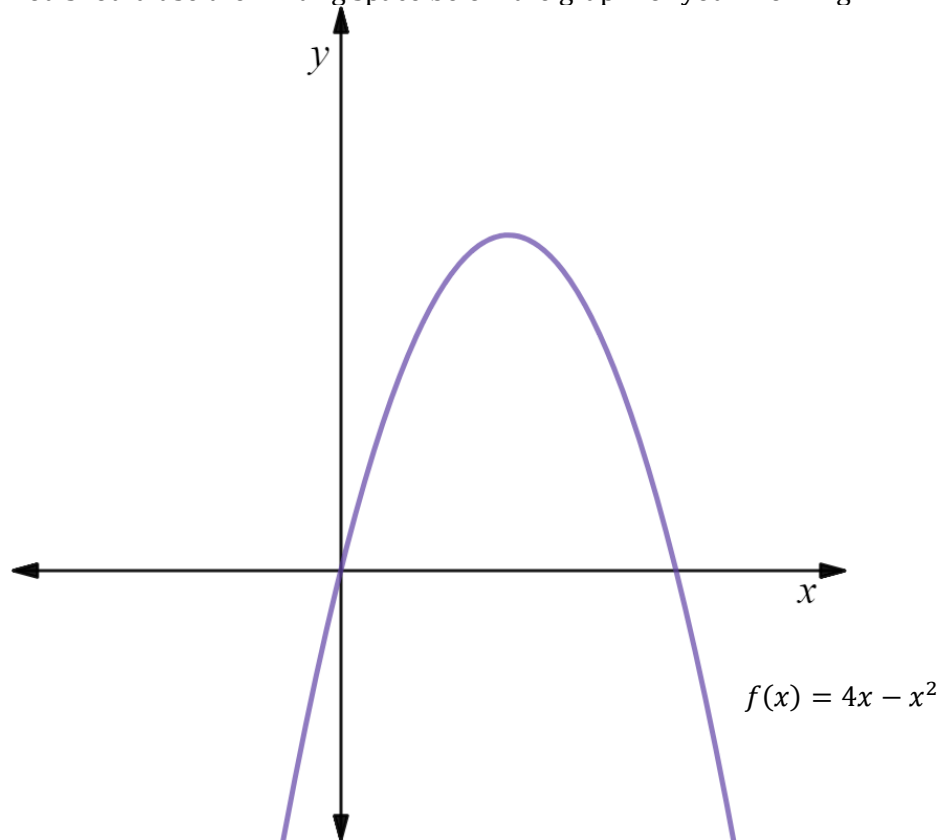
State whether the inverse is a function and **explain** your answer

Question 3 continued

(c) Below is the graph of .

- i) Find the value of all intercepts and the vertex and mark these on the diagram. 1

You should use the writing space below the graph for your working.



- ii) Suggest a restriction on the domain of so that the inverse is a function with maximum possible range. 1

- iii) Sketch the inverse function on the graph above (using your restricted domain from part ii) ensuring that all critical points are clearly marked. 2

Question 3 continued

(d) A function is defined with

i) Show that

1

ii) Hence, show that the inverse and state any restrictions on the range of the inverse function.

2

Question 3 continued

(d) iii) Sketch the graph of clearly marking all critical points.

3

Question 4

F2.1 – Remainder and factor theorems

10 marks

- (a)** Which of the following is the correct solution to the equation ?

1

Circle the correct answer

A.

B.

C.

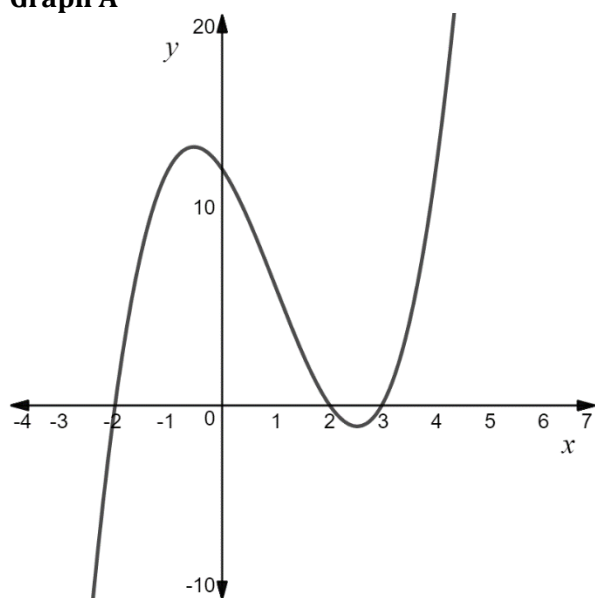
D.

- (b)** A polynomial has as a factor and .
Which of the following is a possible graph of

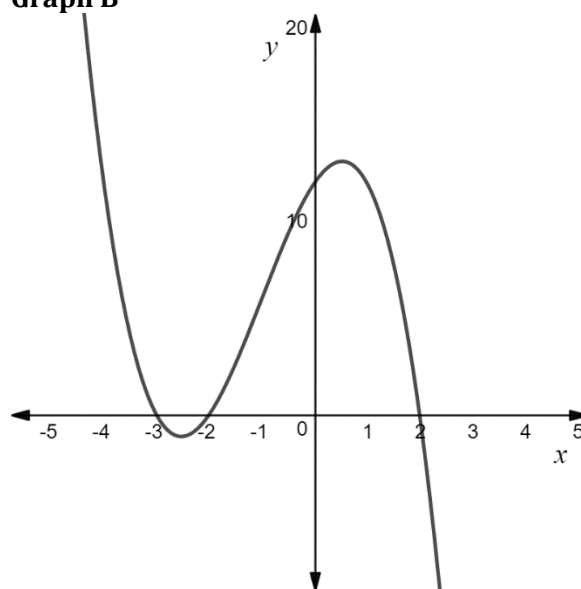
1

Circle the correct graph.

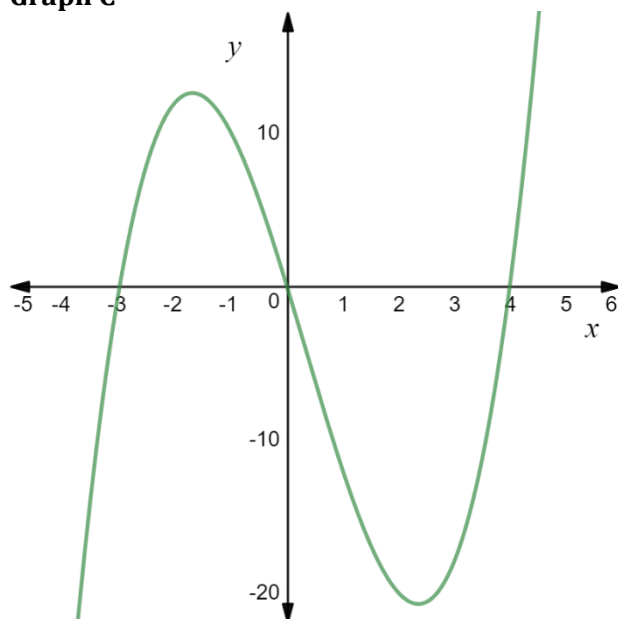
Graph A



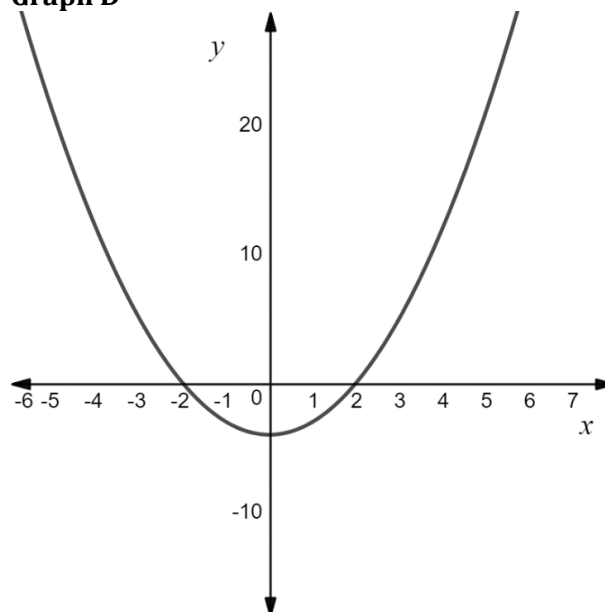
Graph B



Graph C



Graph D



Question 4 continued

(c) Find the remainder when 10^{100} is divided by 101.

1

[illegible]

(d) Show that $x^2 + 1$ is a factor of the polynomial $x^4 + 2x^3 + 3x^2 + 4x + 5$.

1

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 4 continued

(e) Sketch the graph in the blank space below, marking all intercepts. Use the writing space for your working.

4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 4 continued

(f) The polynomials $P(x)$ and $Q(x)$ have a common factor where $d = \deg(P(x)Q(x))$.

Show that

2

END OF TEST

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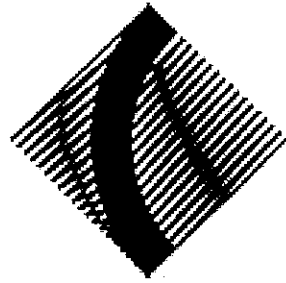
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

NAME: Solutions

CLASS: 11MTX__

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Topic	Total
1 – Graphical relationships	/ 12
2 – Inequalities	/ 10
3 – Inverse functions	/ 11
4 – Factor and remainder theorems	/ 10
Total	/ 43

Question 1

F1.1 Graphical relationships

12 marks

- (a) If $f(x) = |1 - 4x|$ and $g(x) = \sqrt{x}$, what is the domain of $f \circ g(x)$?

Circle the correct answer.

A. all real x

B. $x \geq 0$

C. $x > 0$

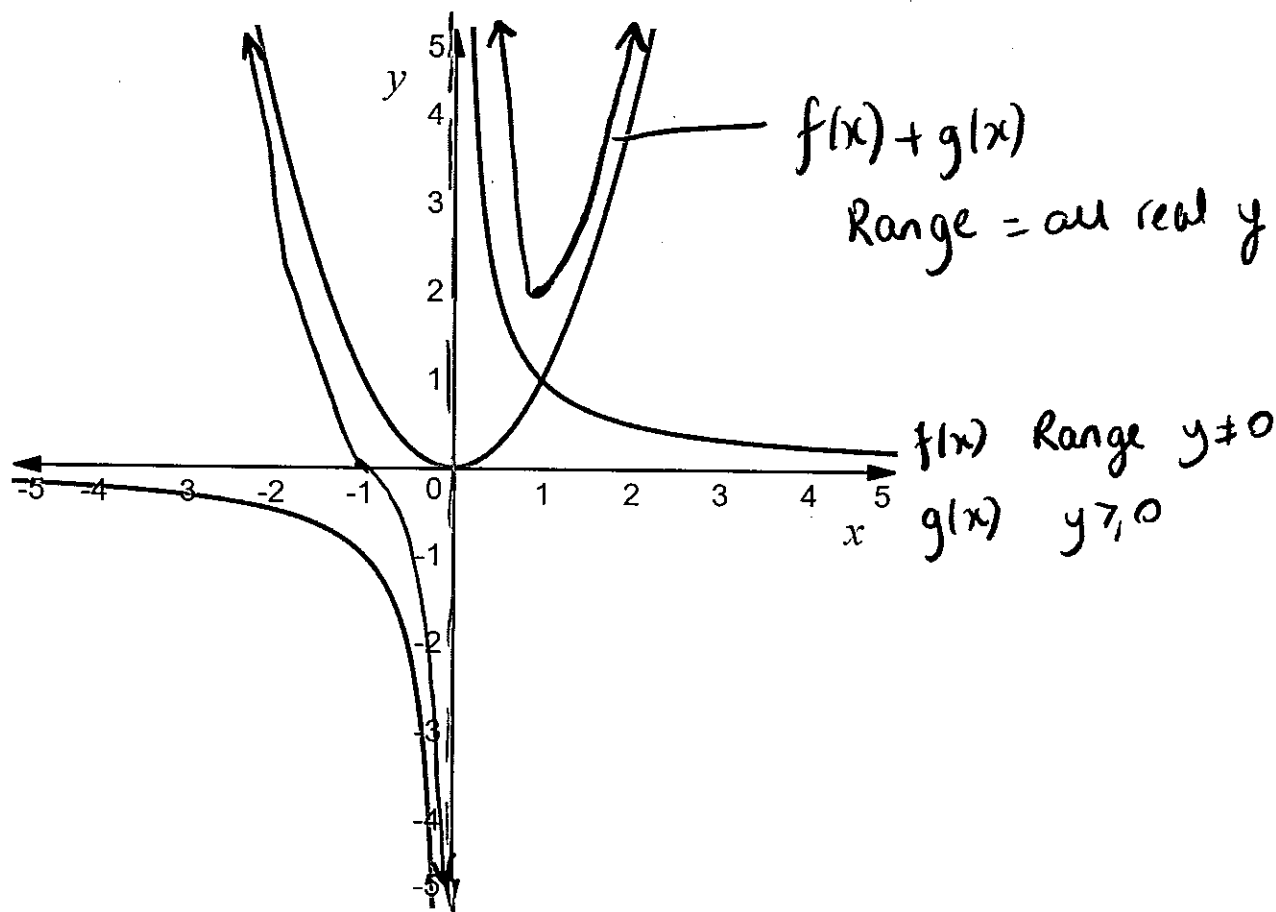
D. $x \geq \frac{1}{4}$

$$f \circ g(x) = |1 - 4\sqrt{x}|$$

↑
only defined for $x \geq 0$

- (b) Below are the graphs of $f(x) = \frac{1}{x}$ and $g(x) = x^2$

1



Circle the correct answer which describes the the **range** of $f(x) + g(x)$.

A. the same as the range of $f(x)$. ✗

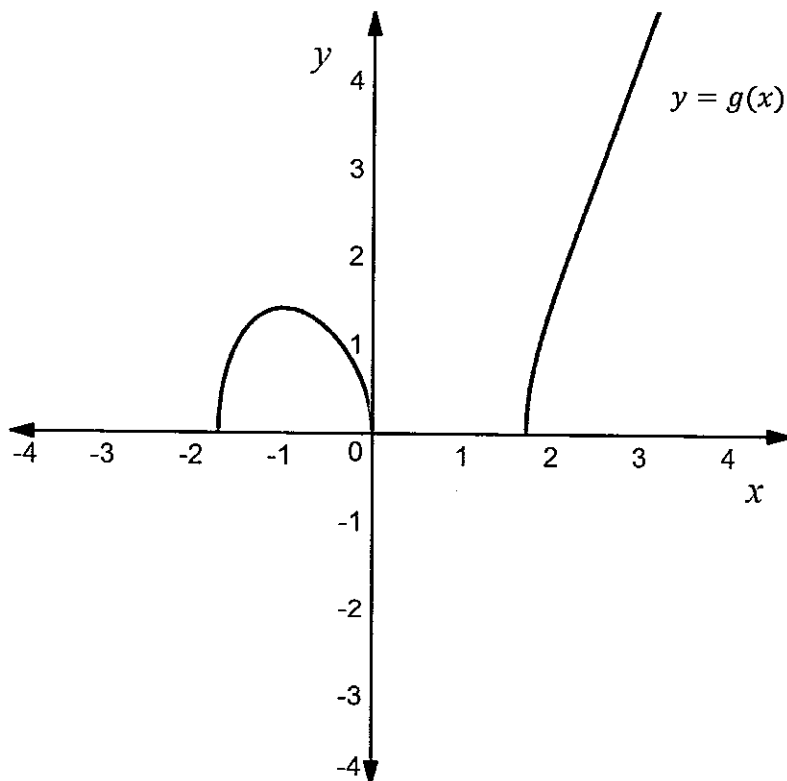
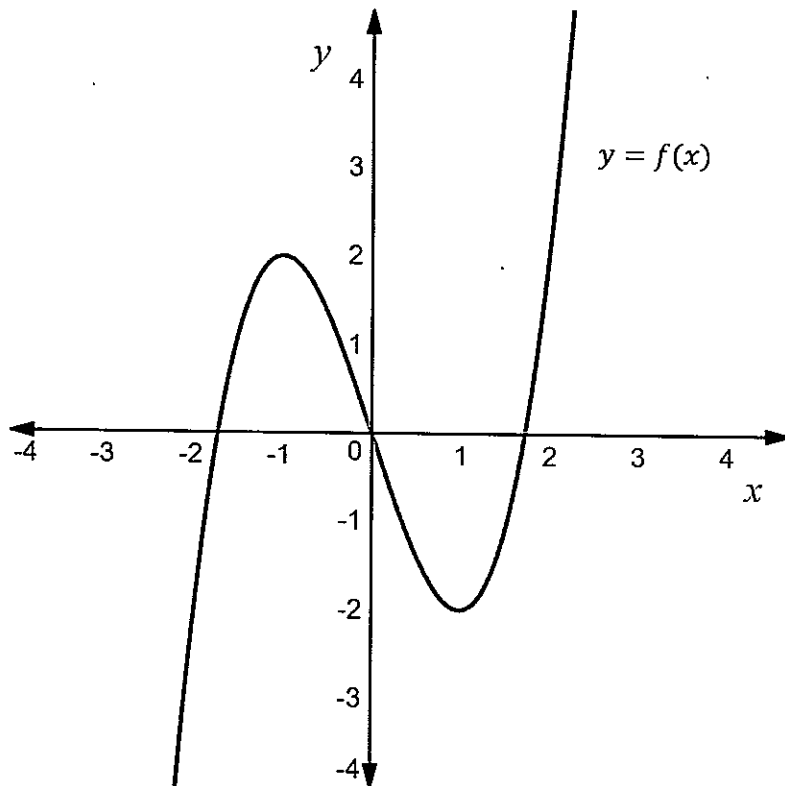
B. the same as the range of $g(x)$. ✗

C. the intersection of the two ranges. ✗

D. None of the above.

Question 1 continued

(c) Below is a graph $y = f(x)$ and a new function $y = g(x)$. Which equation describes $g(x)$?



Circle the correct answer:

A. $g(x) = \sqrt{f(x)}$

B. $g(x) = |f(x)|$

C. $g(x) = f^{-1}(x)$

D. $g(x) = f(|x|)$

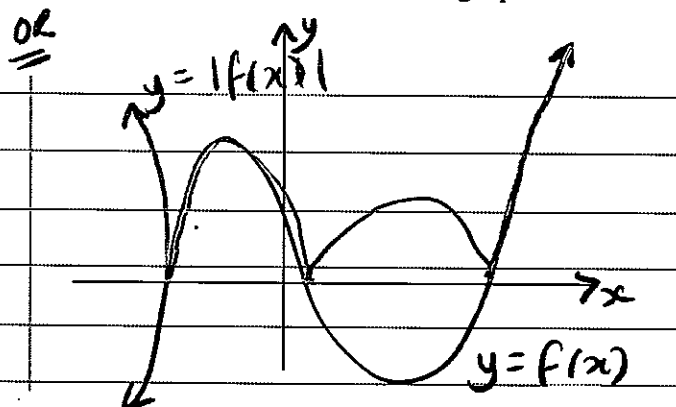
square root function is only defined for $f(x) \geq 0$

Question 1 continued

- (d) Describe in words **OR** illustrate with a suitable diagram, how you would draw the graph of $y = |f(x)|$ if given the graph of $y = f(x)$.

1

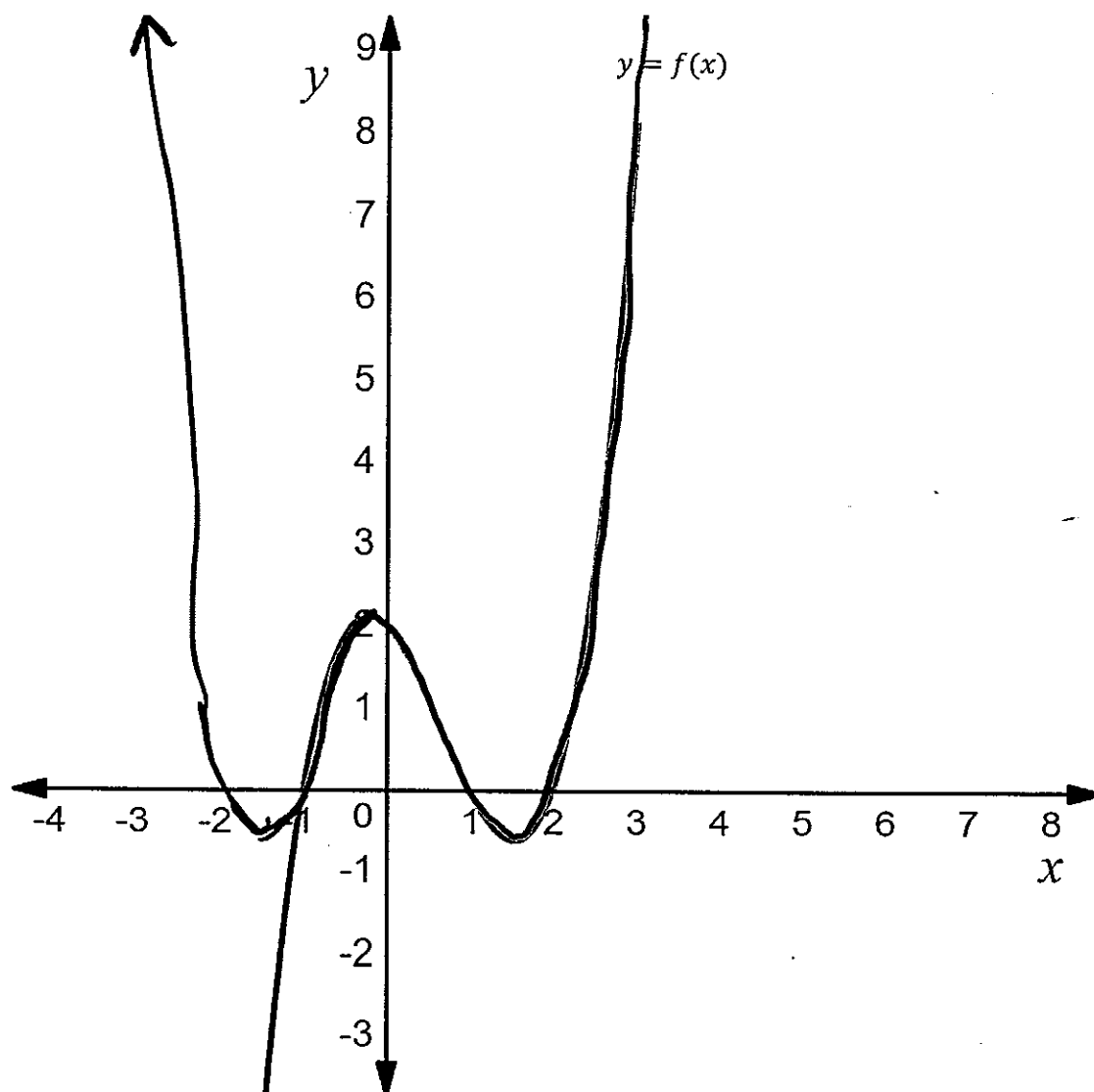
Reflect any parts
below the x -axis
(Q3, Q4) above the
 x -axis



- (e) Below is the graph of a function $y = f(x)$.

2

Sketch the graph $y = f(|x|)$ on the same number plane.

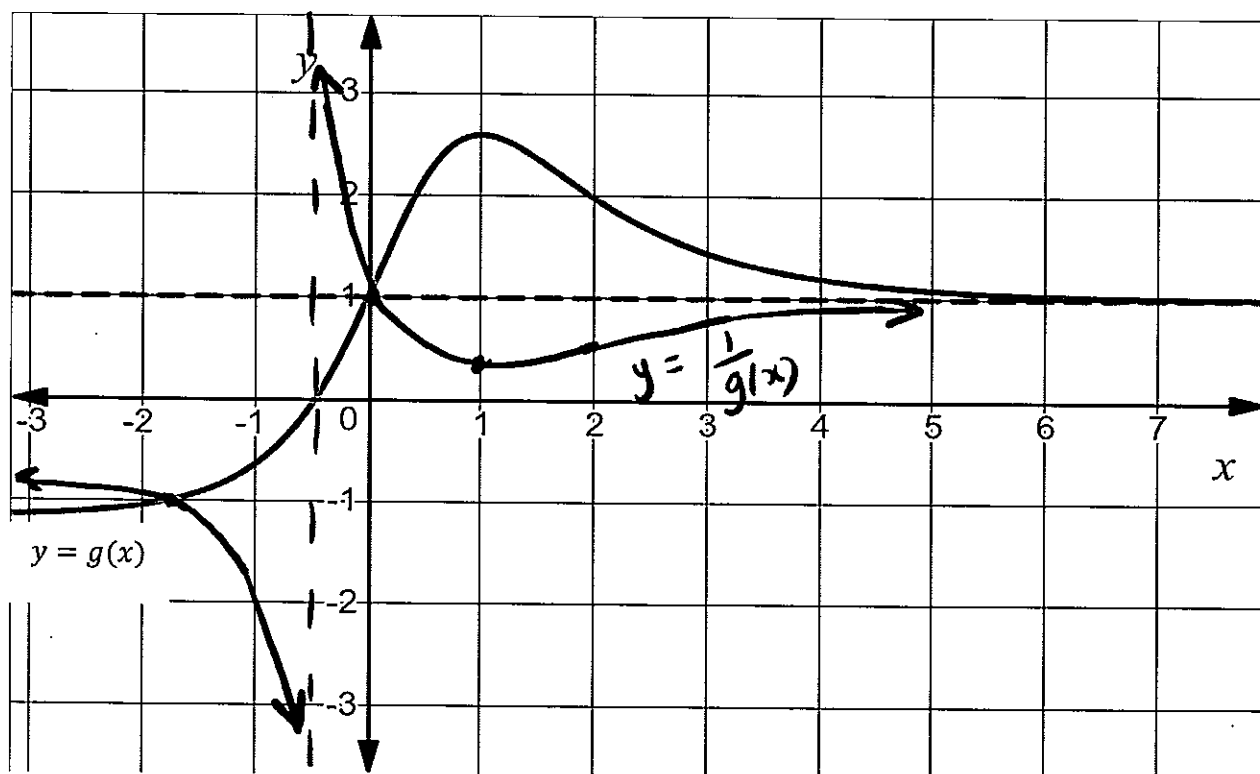


Question 1 continued

- (f) Below is the graph of a function $y = g(x)$.

3

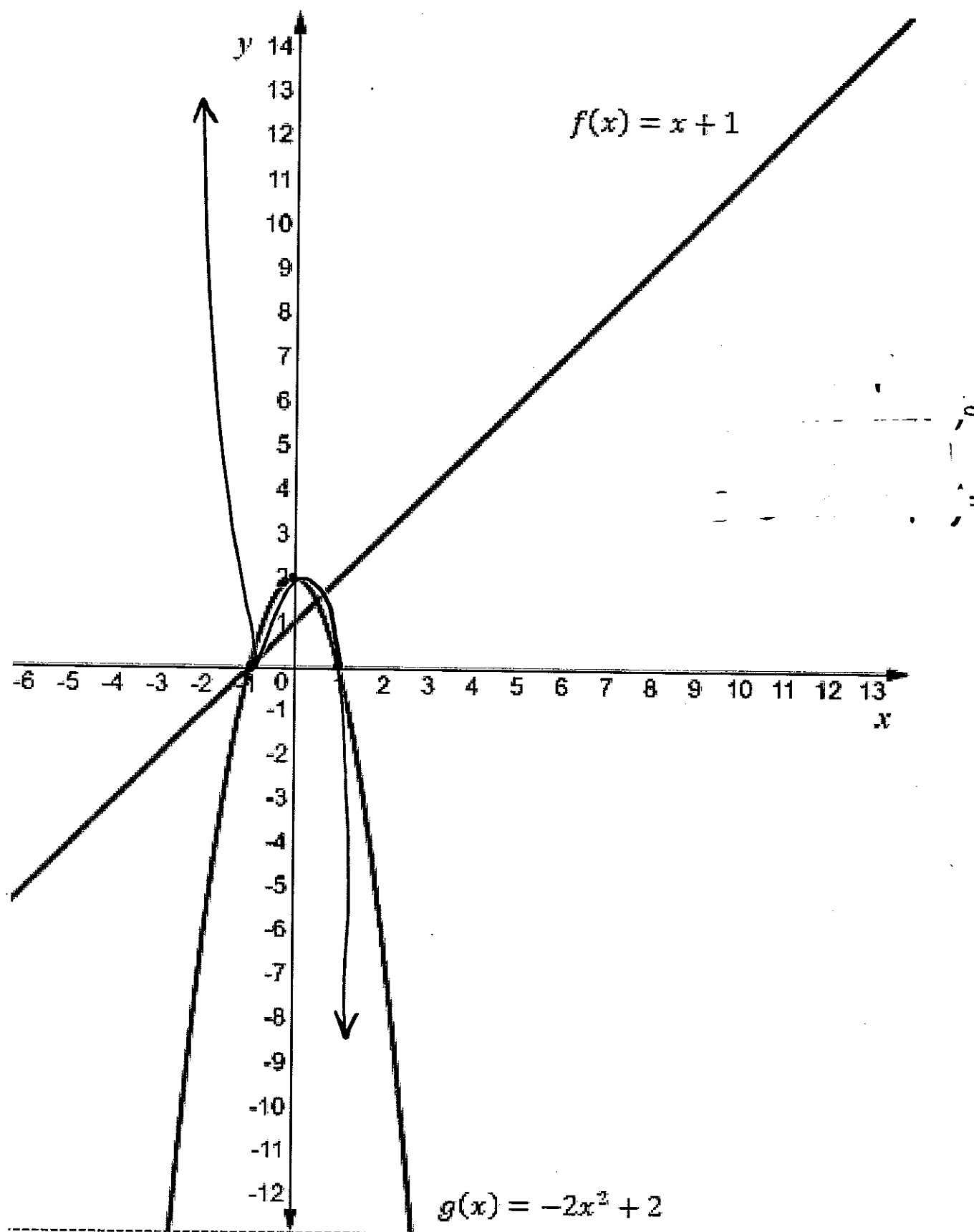
Sketch the graph $y = \frac{1}{g(x)}$ on the same number plane, clearly marking all intercepts and asymptotes.



Question 1 continued

- (g) t** On the number plane below are the graphs of two functions $f(x) = x + 1$ and $g(x) = -2x^2 + 2$. **3**

Sketch the graph of $h(x) = f(x)g(x)$ on the same number plane marking all key features.



Question 2

F1.2 - Inequalities

10 marks

- (a) Solve the inequation $\frac{|-x|}{8} \leq 2$. Circle the correct answer.

1

A. $x \geq 16$

B. $x \leq -16$

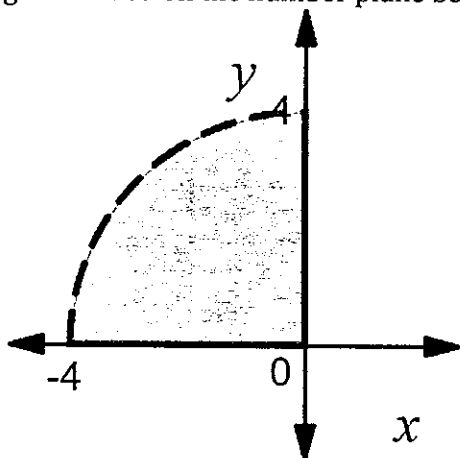
C. $-16 \leq x \leq 16$

D. No solution

$$\begin{aligned} |-x| &\leq 16 \\ |x| &\leq 16 \\ -16 &\leq x \leq 16 \end{aligned}$$

- (b) The region shaded on the number plane below is described by which inequalities?

1



Circle the correct answer

A. $x^2 + y^2 \geq 16$ and $x \leq 0$ or $y \geq 0$

B. $x^2 + y^2 > 4$ and $x \leq 0$ and $y \geq 0$

C. $x^2 + y^2 < 16$ and $x \leq 0$ and $y \geq 0$

D. $x^2 + y^2 \leq 4$ and $x \leq 0$ and $y \geq 0$

- (c) Solve $x^2 - 5x + 6 < 0$.

2

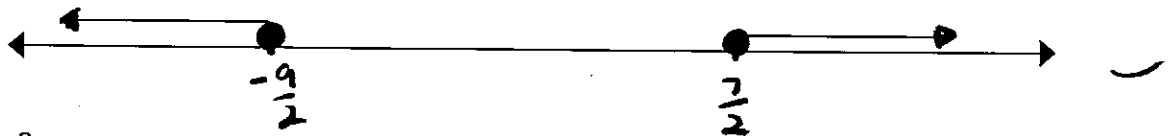
$(x-2)(x-3) < 0$ Roots $x = 2, 3$ 	OR Let $x^2 - 5x + 6 = 0$ $(x-2)(x-3) = 0$ $x = 2, 3$
$2 < x < 3$	$\therefore 2 < x < 3$

Question 2 continued

(d) Solve $|2x + 1| \geq 8$ and sketch your solution on the number line

3

$2x + 1 \geq 8$	$2x + 1 \leq -8$	Let $ 2x + 1 = 8$
$2x \geq 7$	$2x \leq -9$	$2x + 1 = 8, \quad 2x + 1 = -8$
$x \geq \frac{7}{2}$	$x \leq -\frac{9}{2}$	$x = \frac{7}{2}, \quad x = -\frac{9}{2}$
		$x \leq -\frac{9}{2}, \quad x \geq \frac{7}{2}$



(e) Solve $\frac{3x}{x-1} \leq 2$

3

Let $\frac{3x}{x-1} = 2$	$x \neq 1$	OR	$x \neq 1$
$3x = 2(x-1)$			$3x(x-1) \leq 2(x-1)^2$
$3x - 2x + 2 = 0$			$(x-1)(2(x-1) - 3x) \geq 0$
$x = -2$			$(x-1)(-2-x) \geq 0$
			$-(x-1)(2+x) \geq 0$
			Roots: $x = 1, -2$
$-2 \leq x < 1$			
			concave down
			$-2 \leq x < 1$

- (a) If $f(x) = \frac{2}{3}x - 5$ find the equation of the inverse function $f^{-1}(x)$.

1

$$f(x): y = \frac{2}{3}x - 5$$

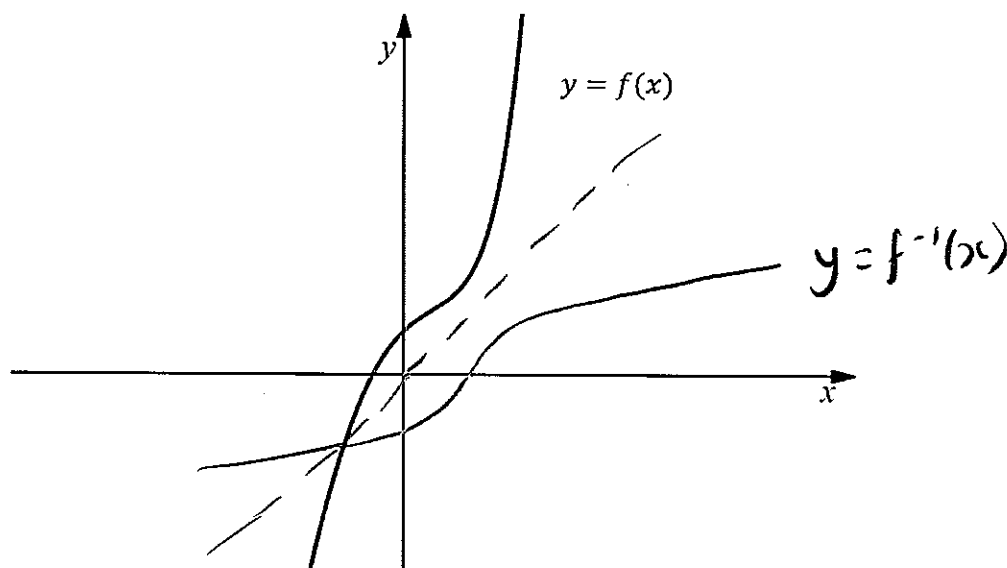
$$f^{-1}(x): x = \frac{2}{3}y - 5$$

$$x + 5 = \frac{2}{3}y$$

$$y = \frac{3}{2}(x + 5) \quad \text{OR} \quad y = \frac{3x}{2} + \frac{15}{2}$$

- (b) Below is the graph of a function $y = f(x)$.

1



State whether the inverse $f^{-1}(x)$ is a function and **explain** your answer

$f(x)$ is a function as $f(x)$ passes the horizontal line test.

OR $f^{-1}(x)$ is a function as it passes the vertical line test $f^{-1}(x)$ must be drawn on graph as above

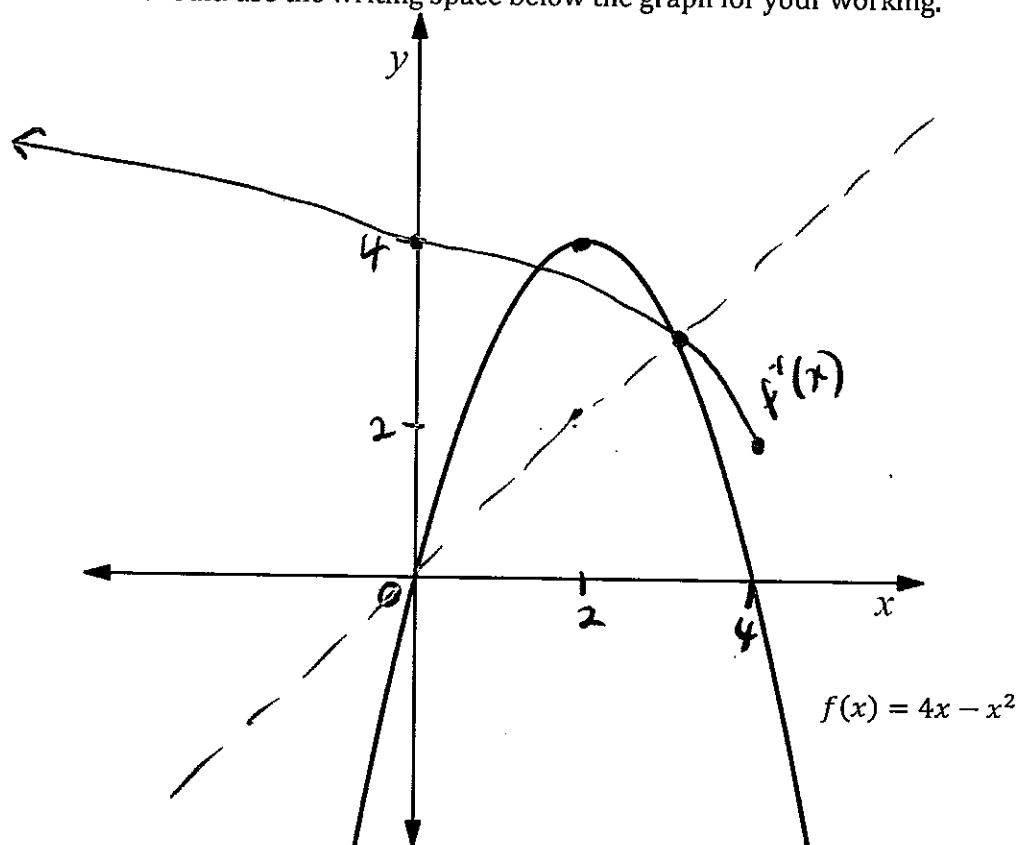
Question 3 continued

(c) Below is the graph of $f(x) = 4x - x^2$.

i) Find the value of all intercepts and the vertex and mark these on the diagram.

1

You should use the writing space below the graph for your working.



$$x\text{-int: } 4x - x^2 = 0$$

$$x(4 - x) = 0$$

$$x = 0, 4$$

$$\text{vertex } x = \frac{0+4}{2}$$

$$= 2$$

$$\begin{aligned} y &= 4(2) - 2^2 \\ &= 8 - 4 \\ &= 4 \end{aligned}$$

ii) Suggest a restriction on the domain of $f(x)$ so that the inverse $f^{-1}(x)$ is a function with maximum possible range.

1

$$x \geq 2 \quad \text{OR} \quad x \leq 2$$

iii) Sketch the inverse function $f^{-1}(x)$ on the graph above (using your restricted domain from part ii) ensuring that all critical points are clearly marked.

2

Question 3 continued

(d) A function is defined $f(x) = \frac{5}{(x+1)(2x-3)} + \frac{1}{x+1}$, $x \neq -1, x \neq \frac{3}{2}$.

i) Show that $\frac{5}{(x+1)(2x-3)} + \frac{1}{x+1} = \frac{2}{2x-3}$

1

$$\text{LHS} = \frac{5}{(x+1)(2x-3)} + \frac{1}{x+1}$$

$$= \frac{5 + (2x-3)}{(x+1)(2x-3)}$$

$$= \frac{2(\cancel{x+1})}{(\cancel{x+1})(2x-3)}$$

$$= \frac{2}{2x-3}$$

SHOW THAT

ii) Hence, show that the inverse $f^{-1}(x) = \frac{1}{x} + \frac{3}{2}$ and state any restrictions on the range of the function. *inverse.*

2

$$f(x) : y = \frac{2}{2x-3}$$

$$f^{-1}(x) : x = \frac{2}{2y-3}$$

$$2y-3 = \frac{2}{x}$$

$$y = \frac{1}{2} \left(\frac{2}{x} + 3 \right)$$

$$y = \frac{1}{x} + \frac{3}{2}$$

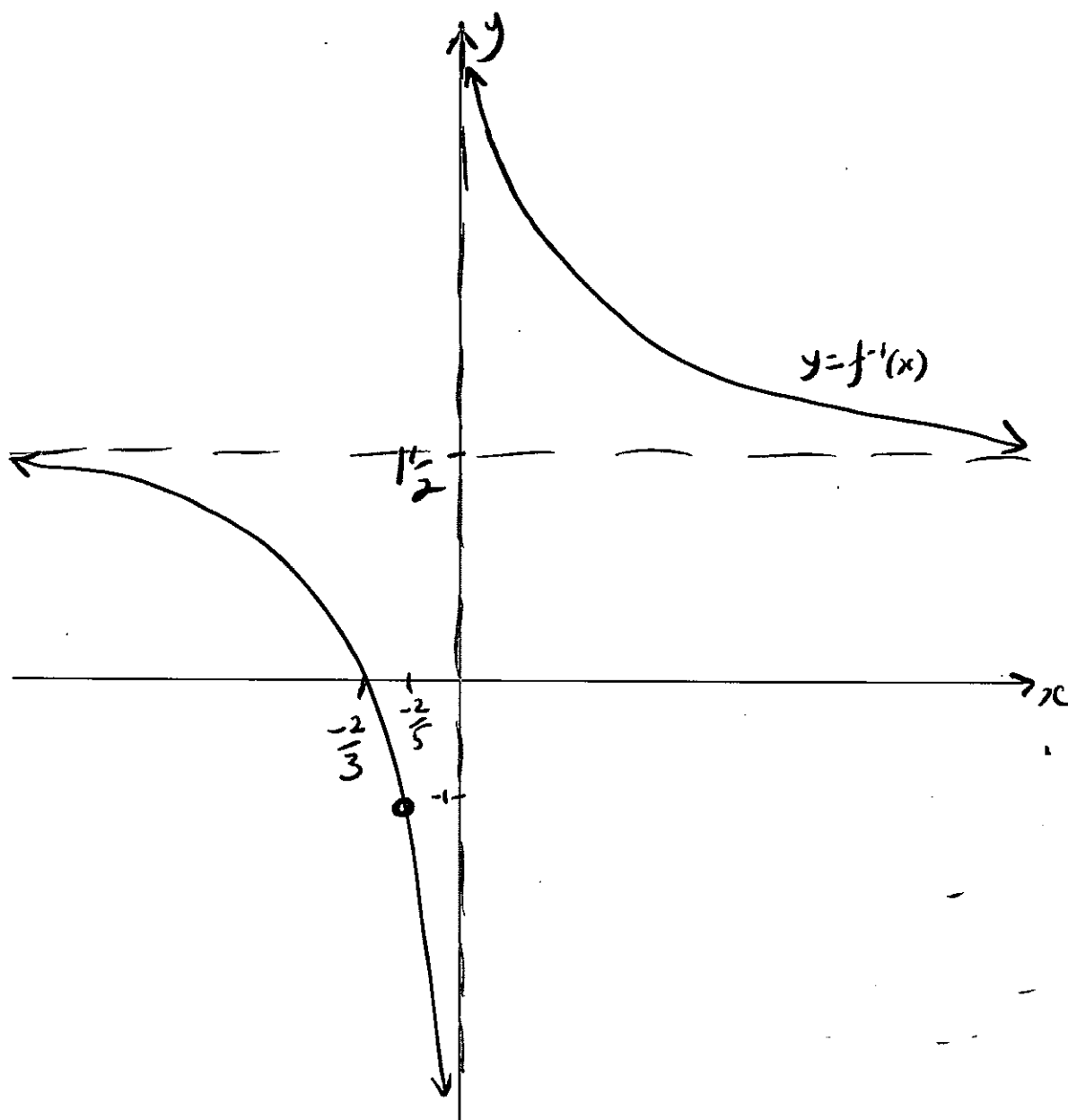
SHOW THAT

Restrictions on range $y \neq -1, y \neq \frac{3}{2}$

Question 3 continued

- (d) iii) Sketch the graph of $f^{-1}(x)$ clearly marking all critical points.

23



$$x\text{-intercept: } y = 0 \quad \frac{1}{x} + \frac{3}{2} = 0 \quad \frac{1}{x} = -\frac{3}{2} \quad x = -\frac{2}{3}$$

$$\text{undefined at } y = -1 \quad \frac{1}{x} + \frac{3}{2} = -1 \quad \frac{1}{x} = -\frac{5}{2} \quad x = -\frac{2}{5}$$

Question 4

F2.1 – Remainder and factor theorems

10 marks

- (a) Which of the following is the correct solution to the equation $x^3 - 3x^2 + 4 = 0$?

1

Circle the correct answer

A. $x = -1$

B. $x = -1, 2$

C. $x = 1, -2$

D. $x = 1, -2, 2$

$P(-1) = 0 \therefore x+1$ is a factor

$$x^3 - 3x^2 + 4 = (x+1)(x^2 - 4x + 4)$$

$$= (x+1)(x-2)^2$$

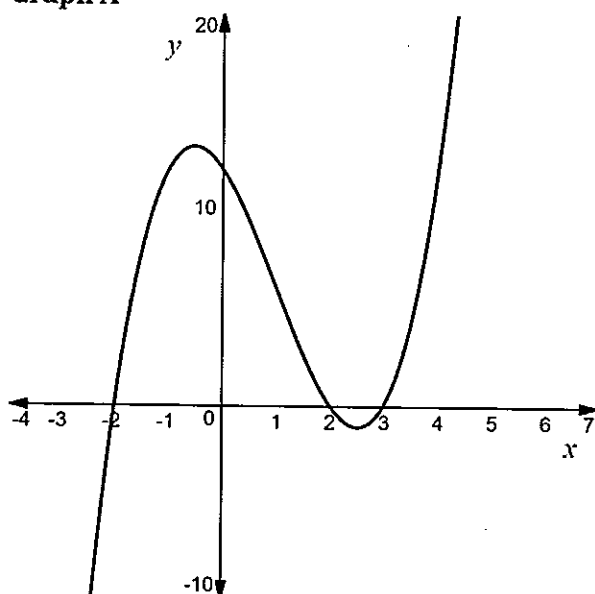
$\therefore$ roots $-1, 2$

- (b) A polynomial $P(x)$ has $x^2 - 4$ as a factor and $P(-3) = 0$. Which of the following is a possible graph of $P(x)$?

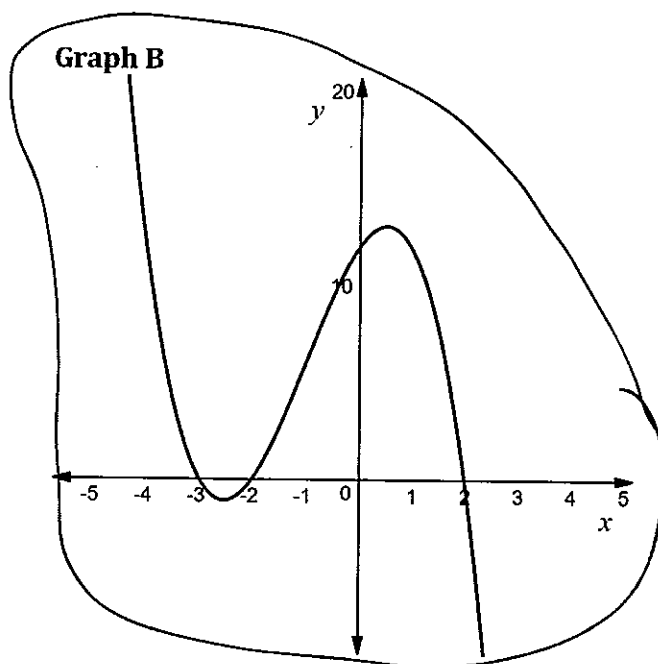
$(x-2)(x+2)(x+3)$ are factors
 $\therefore$ Roots $2, -2, -3$

Circle the correct graph.

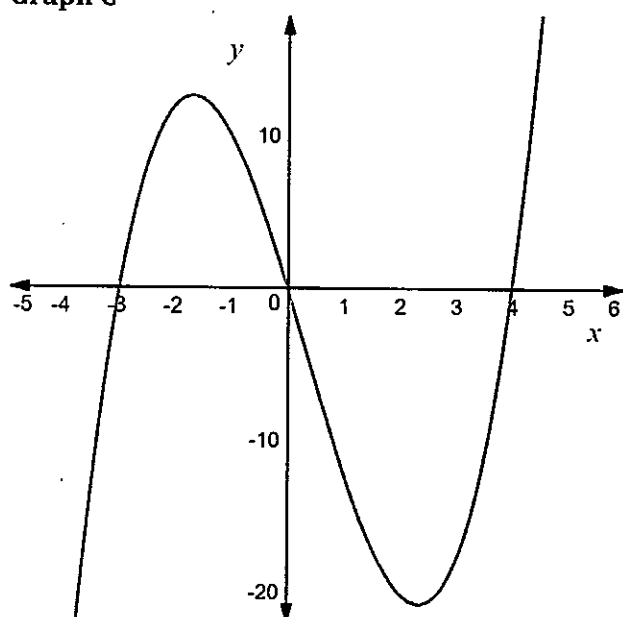
Graph A



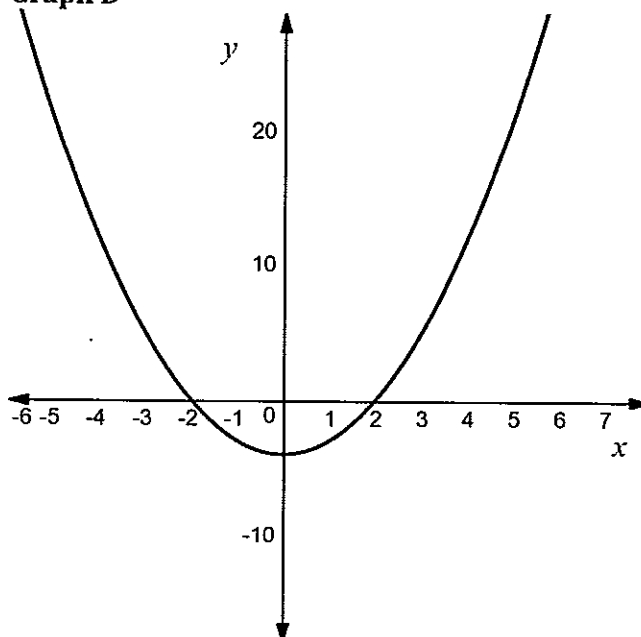
Graph B



Graph C



Graph D



1

Question 4 continued

- (c) Find the remainder when $x^3 - x^2 + x - 4$ is divided by $x - 3$.

1

$P(x) = x^3 - x^2 + x - 4$	$x^2 + 2x + 7$
$P(3) = 3^3 - 3^2 + 3 - 4$	$x-3 \overline{) x^3 - x^2 + x - 4}$
$= 27 - 9 + 3 - 4$	$\underline{- x^3 + 3x^2}$
$= 17$	$2x^2 + x$
	$\underline{2x^2 - 6x}$
$\therefore \text{Remainder} = 17$	$7x - 4$
	$\underline{- 7x + 21}$
	17

- (d) Show that $x + 1$ is a factor of the polynomial $2x^3 - x^2 - 4x - 1$.

1

$P(x) = 2x^3 - x^2 - 4x - 1$	$2x^2 - 3x - 1$
$P(-1) = 2(-1)^3 - (-1)^2 - 4(-1) - 1$	$x+1 \overline{) 2x^3 - x^2 - 4x - 1}$
$= -2 - 1 + 4 - 1$	$\underline{- 2x^3 + 2x^2}$
$= 0$	$-3x^2 - 4x$
	$\underline{- 3x^2 - 3x}$
	$-x - 1$
$P(-1) = 0 \therefore x+1$ is a factor	$\underline{- -x + 1}$
	0
	$R(x) = 0 \therefore x+1$ is a factor

Question 4 continued

- (e) Sketch the graph $y = 2x^3 - 5x^2 - 9x + 18$ in the blank space below, marking all intercepts.

4

Use the writing space for your working.

$$p(x) = 2x^3 - 5x^2 - 9x + 18$$

$$p(-2) = 2(-2)^3 - 5(-2)^2 - 9(-2) + 18$$

$$= -16 - 20 + 18 + 18$$

$$= 0$$

$\therefore x+2$ is a factor

$$2x^2 - 9x + 9$$

$$x+2 \overline{) 2x^3 - 5x^2 - 9x + 18}$$

$$2x^3 + 4x^2$$

$$-9x^2 - 9x$$

$$-9x^2 - 18x$$

$$9x + 18$$

$$9x + 18$$

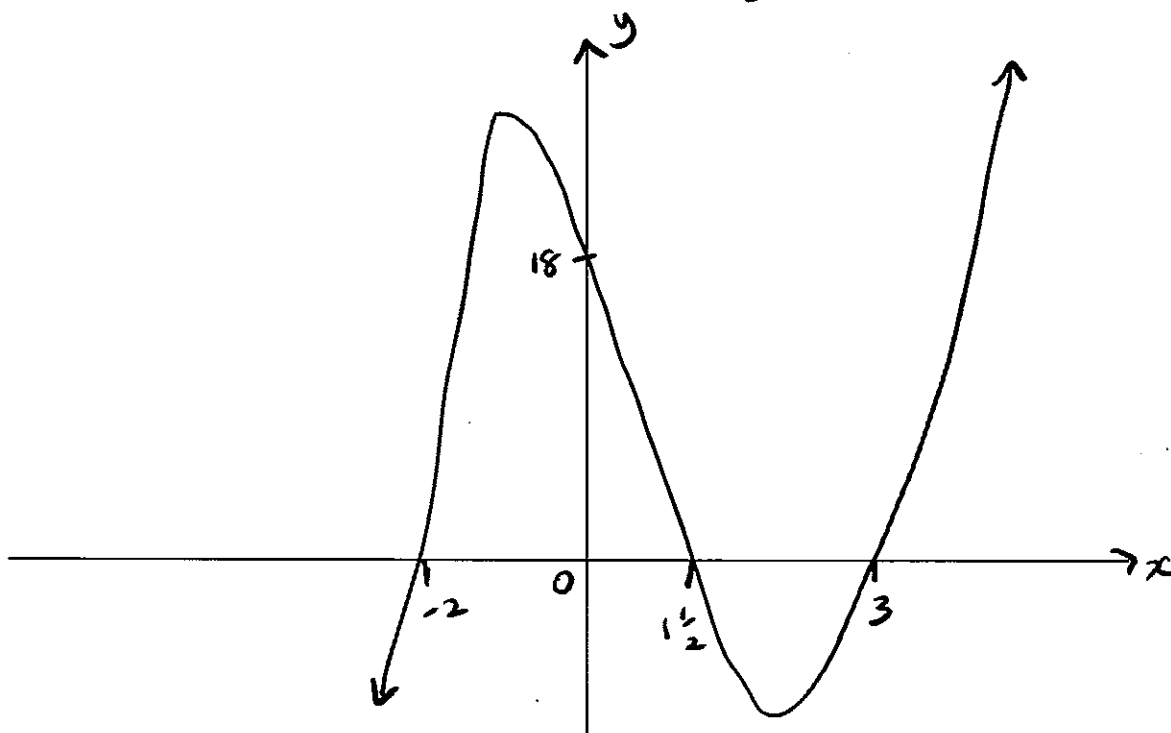
$$0$$

$$2x^3 - 5x^2 - 9x + 18 = (x+2)(2x^2 - 9x + 9)$$

$$= (x+2)(2x-3)(x-3)$$

$\therefore$ Roots $x = -2, \frac{3}{2}, 3$

y-intercept = 18



Question 4 continued

- (f) The polynomials $P(x) = x^2 - ax - b$ and $Q(x) = 2x^2 + b$ have a common factor $x + c$ where $a, b, c \neq 0$.

Show that $b = \frac{2ac}{3}$.

2

$$P(-c) = 0$$

$$Q(c) = 0$$

$$P(-c) = (-c)^2 - a(-c) - b$$

$$Q(c) = 2(c)^2 + b$$

$$\therefore c^2 + ac - b = 0 \quad (1)$$

$$\therefore 2c^2 + b = 0 \quad (2)$$

$$\text{From (2) } c^2 = -\frac{b}{2}$$

sub into (1)

$$-\frac{b}{2} + ac - b = 0$$

$$-\frac{3b}{2} + ac = 0$$

$$\frac{3b}{2} = ac$$

$$b = \frac{2ac}{3}$$

SHOW
THAT

END OF TEST